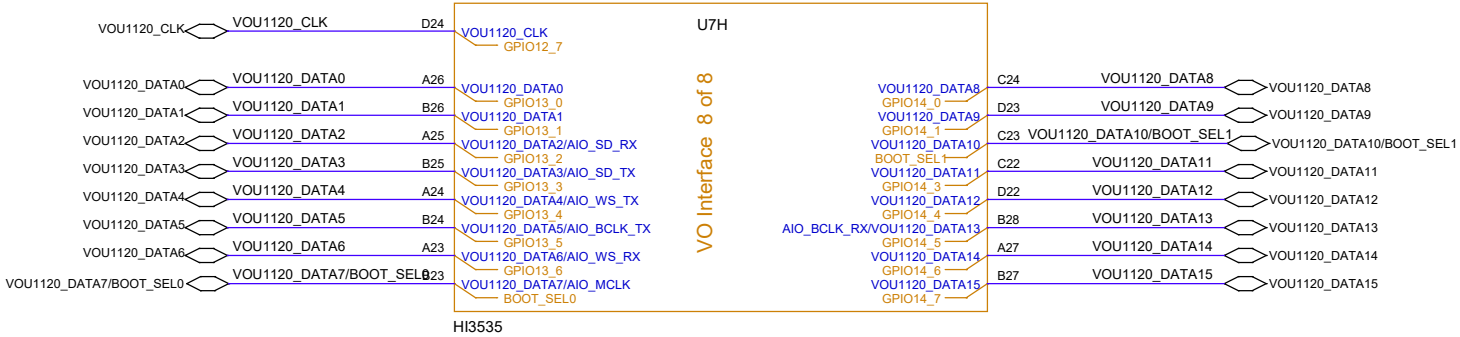
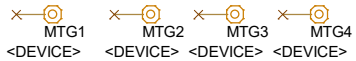
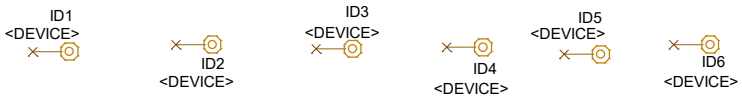
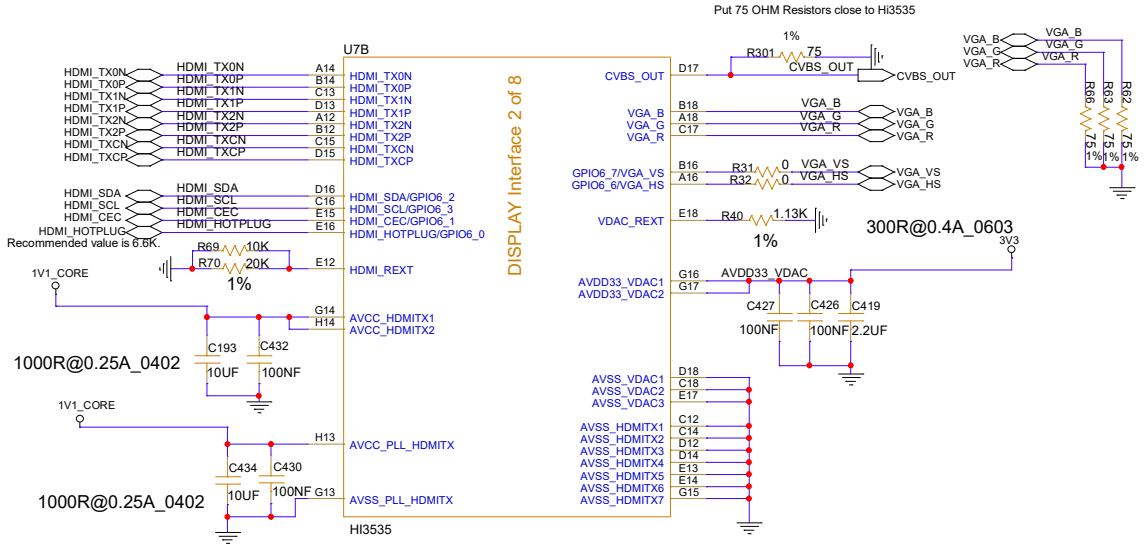
VO

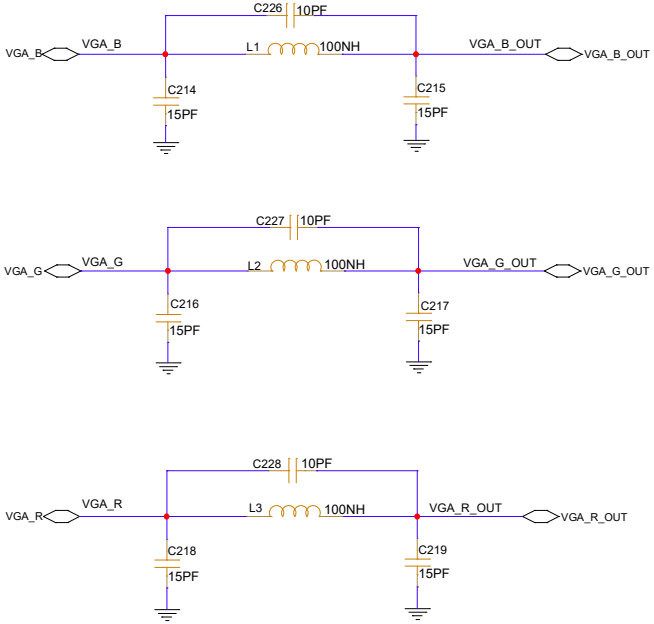

U7H
VO Interface 8 of 8
HI3535
VOU1120_CLK
D24
VOU1120_CLK
GPIO12_7
VOU1120_DATA0
A26
VOU1120_DATA0
GPIO13_0
VOU1120_DATA1
B26
VOU1120_DATA1
GPIO13_1
VOU1120_DATA2
A25
VOU1120_DATA2/AIO_SD_RX
GPIO13_2
VOU1120_DATA3
B25
VOU1120_DATA3/AIO_SD_TX
GPIO13_3
VOU1120_DATA4
A24
VOU1120_DATA4/AIO_WS_TX
GPIO13_4
VOU1120_DATA5
B24
VOU1120_DATA5/AIO_BCLK_TX
GPIO13_5
VOU1120_DATA6
A23
VOU1120_DATA6/AIO_WS_RX
GPIO13_6
VOU1120_DATA7/BOOT_SEL0
B23
VOU1120_DATA7/AIO_MCLK
BOOT_SEL0
VOU1120_DATA8
C24
VOU1120_DATA8
GPIO14_0
VOU1120_DATA9
D23
VOU1120_DATA9
GPIO14_1
VOU1120_DATA10/BOOT_SEL1
C23
VOU1120_DATA10
BOOT_SEL1
VOU1120_DATA11
C22
VOU1120_DATA11
GPIO14_3
VOU1120_DATA12
D22
VOU1120_DATA12
GPIO14_4
AIO_BCLK_RX/VOU1120_DATA13
B28
VOU1120_DATA13
GPIO14_5
VOU1120_DATA14
A27
VOU1120_DATA14
GPIO14_6
VOU1120_DATA15
B27
VOU1120_DATA15
GPIO14_7


MTG1
<DEVICE>
MTG2
<DEVICE>
MTG3
<DEVICE>
MTG4
<DEVICE>


ID1
<DEVICE>
ID2
<DEVICE>
ID3
<DEVICE>
ID4
<DEVICE>
ID5
<DEVICE>
ID6
<DEVICE>

DISPLAY


Put 75 OHM Resistors close to Hi3535
U7B
DISPLAY Interface 2 of 8
HI3535
HDMI_TX0N
HDMI_TX0P
HDMI_TX1N
HDMI_TX1P
HDMI_TX2N
HDMI_TX2P
HDMI_TXCN
HDMI_TXCP
HDMI_SDA
HDMI_SCL
HDMI_CEC
HDMI_HOTPLUG
Recommended value is 6.6K.
R69 10K
R70 20K
1%
1V1_CORE
HDMI_REXT
HDMI_SDA/GPIO6_2
HDMI_SCL/GPIO6_3
HDMI_CEC/GPIO6_1
HDMI_HOTPLUG/GPIO6_0
AVCC_HDMITX1
AVCC_HDMITX2
AVCC_PLL_HDMITX
AVSS_PLL_HDMITX
C193 10UF
C432 100NF
1000R@0.25A_0402
C434 10UF
C430 100NF
CVBS_OUT
R301 75 1%
VGA_B
VGA_G
VGA_R
GPIO6_7/VGA_VS
GPIO6_6/VGA_HS
VGA_VS
VGA_HS
R31 0
R32 0
VDAC_REXT
R40 1.13K 1%
300R@0.4A_0603
3V3
AVDD33_VDAC1
AVDD33_VDAC2
AVDD33_VDAC
C427 100NF
C426 100NF
C419 2.2UF
AVSS_VDAC1
AVSS_VDAC2
AVSS_VDAC3
AVSS_HDMITX1
AVSS_HDMITX2
AVSS_HDMITX3
AVSS_HDMITX4
AVSS_HDMITX5
AVSS_HDMITX6
AVSS_HDMITX7
R66 75 1%
R63 75 1%
R62 75 1%

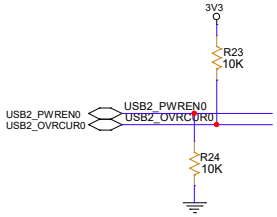

C226 10PF
VGA_B
L1 100NH
VGA_B_OUT
C214 15PF
C215 15PF
C227 10PF
VGA_G
L2 100NH
VGA_G_OUT
C216 15PF
C217 15PF
C228 10PF
VGA_R
L3 100NH
VGA_R_OUT
C218 15PF
C219 15PF

USB & SATA

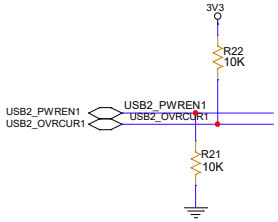
The SATA differential trace impedance is 100 OHM.

The SATA trace length is less than 5 inch.

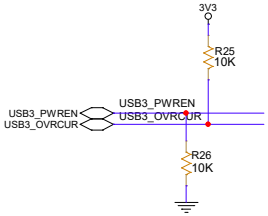
USB2.0 Port0



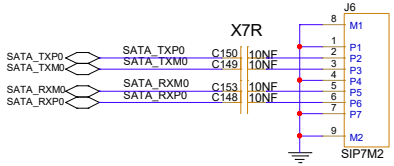
USB2.0 Port1



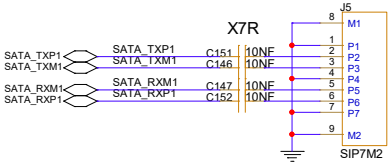
USB3.0



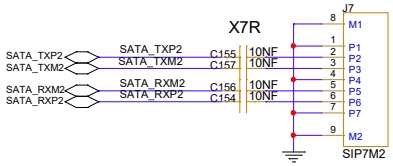
SATA3.0 Port0

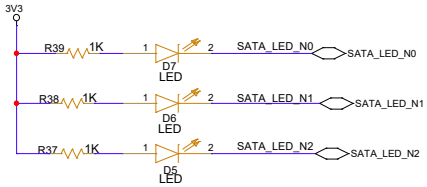


SATA3.0 Port1



SATA3.0 Port2

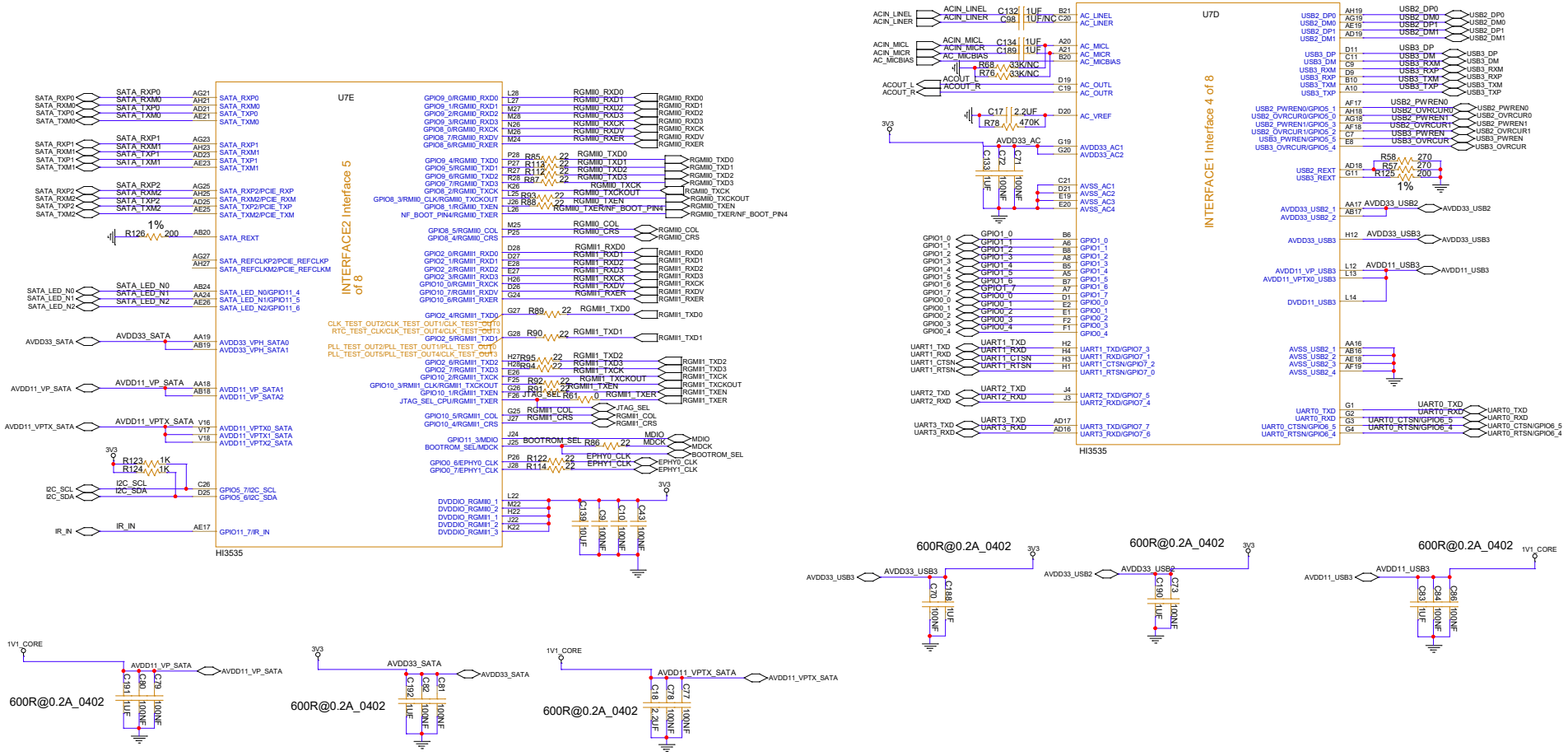




The USB differential trace impedance is 90 OHM.

The USB trace length is less than 5 inch.

Interface


U7E
INTERFACE2 Interface 5 of 8
HI3535
U7D
INTERFACE1 Interface 4 of 8
HI3535
600R@0.2A_0402
600R@0.2A_0402
600R@0.2A_0402
600R@0.2A_0402
600R@0.2A_0402
600R@0.2A_0402

Power on setting pins

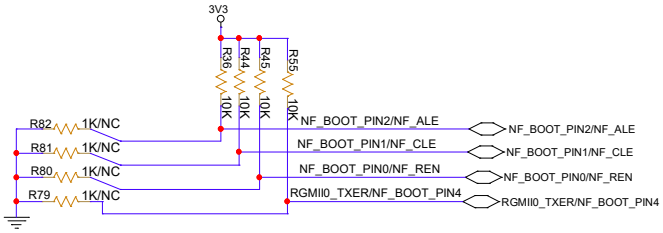
NF_BOOT_PIN[4:0]

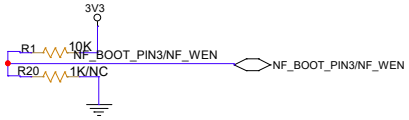
00011 2k page size,4bit ecc,64page/block,5addr

00100 4k page size,24bit ecc,256page/block,5addr
00101 2k page size,24bit ecc,64page/block,5addr
00111 8k page size,24bit ecc,256page/block,5addr

01000 4k page size,4bit ecc,128page/block,5addr
01001 4k page size,4bit ecc,64page/block,5addr
01010 2k page size,4bit ecc,64page/block,4addr
01011 4k page size,24bit ecc,128page/block,5addr
01101 8k page size,24bit ecc,128page/block,5addr

10000 8k page size,24bit ecc,64page/block,5addr
10001 4k page size,24bit ecc,64page/block,5addr
10101 2k page size,4bit ecc,128page/block,5addr
10110 2k page size,8bit ecc,128page/block,5addr
11001 2k page size,24bit ecc,128page/block,5addr
11010 2k page size,8bit ecc,64page/block,5addr
11110 4k page size,8bit ecc,64page/block,5addr
11111 4k page size,8bit ecc,128page/block,5addr

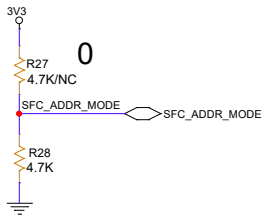


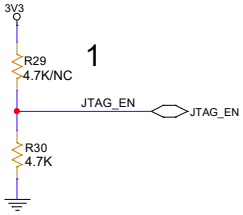


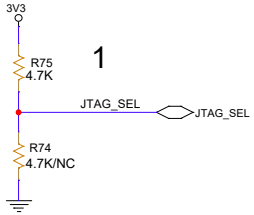
	SFC_ADDR_MODE
0	3 Byte mode
1	4 Byte mode

	JTAG_EN
0	Disable JTAG
1	Enable JTAG

	JTAG_SEL
0	Other
1	CPU

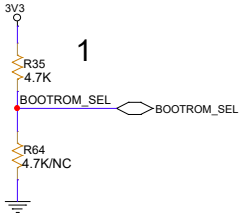


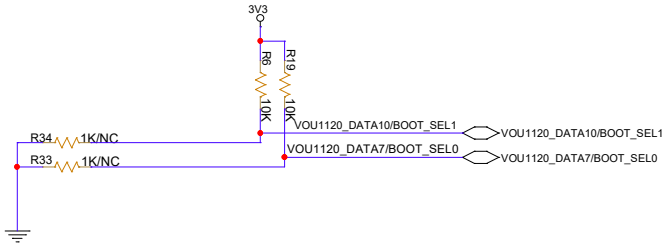




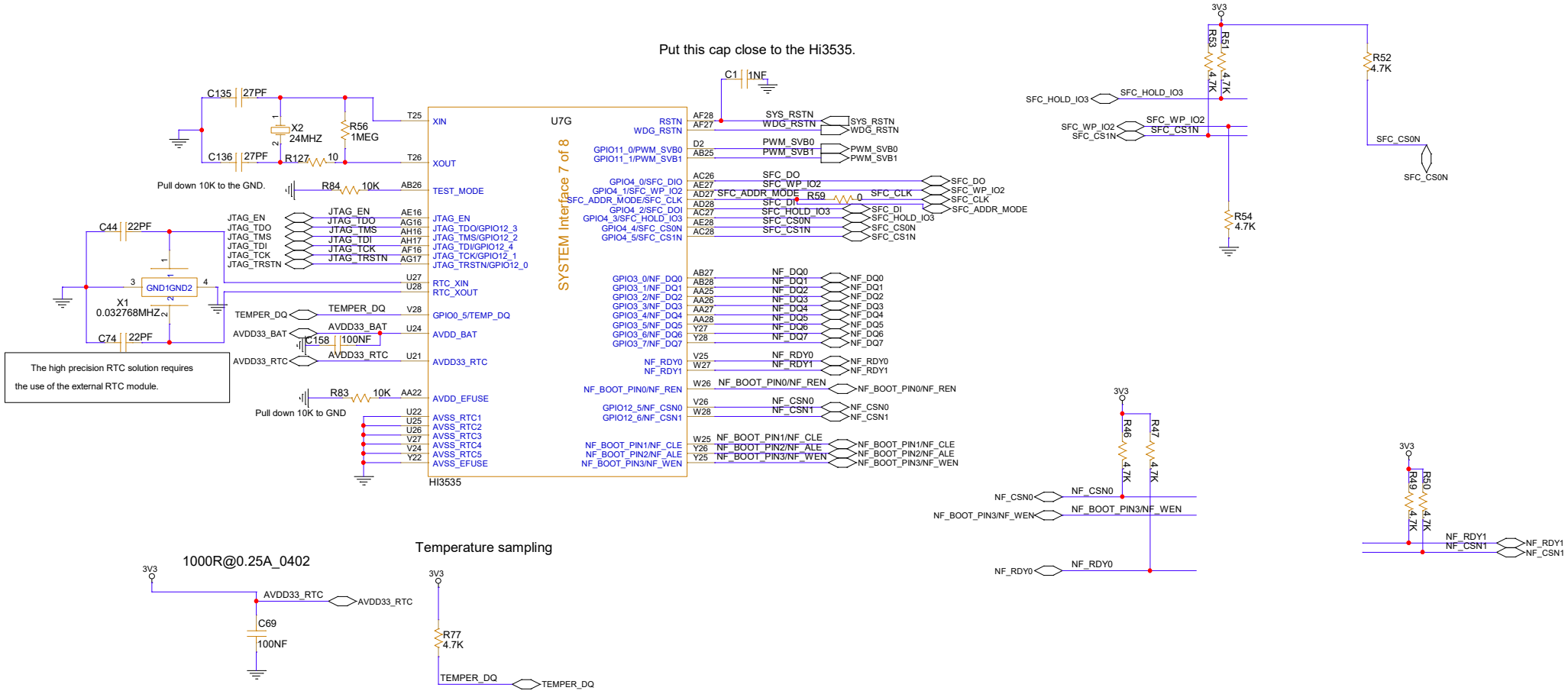
	BOOTROM_SEL
0	Boot from BOOT_SEL
1	Boot from Bootrom

	BOOT_SEL[1:0]
00	SPI FLASH
01	DDR
10	NAND FLASH
11	Reserve





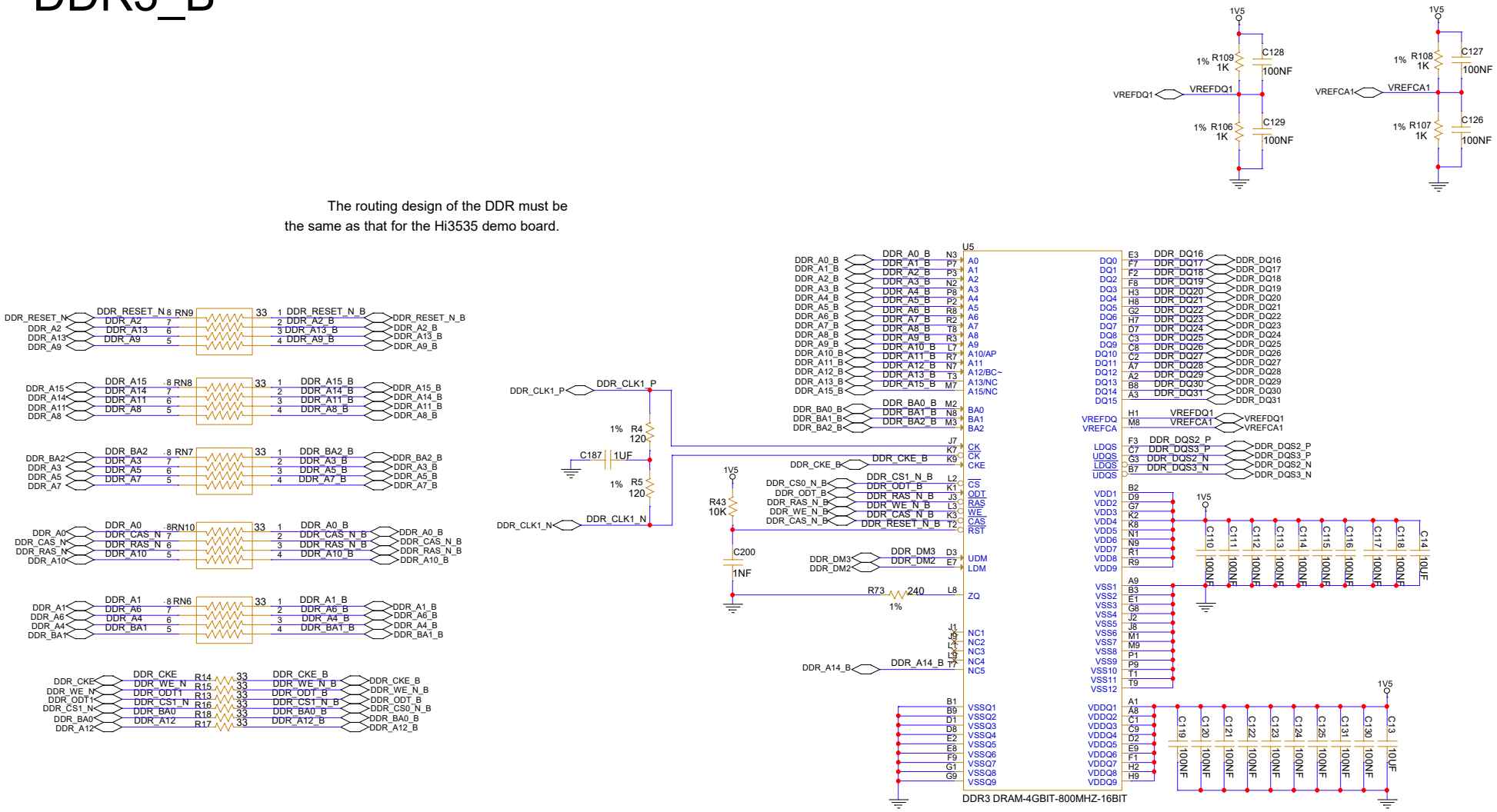
SYSTEM & FLASH



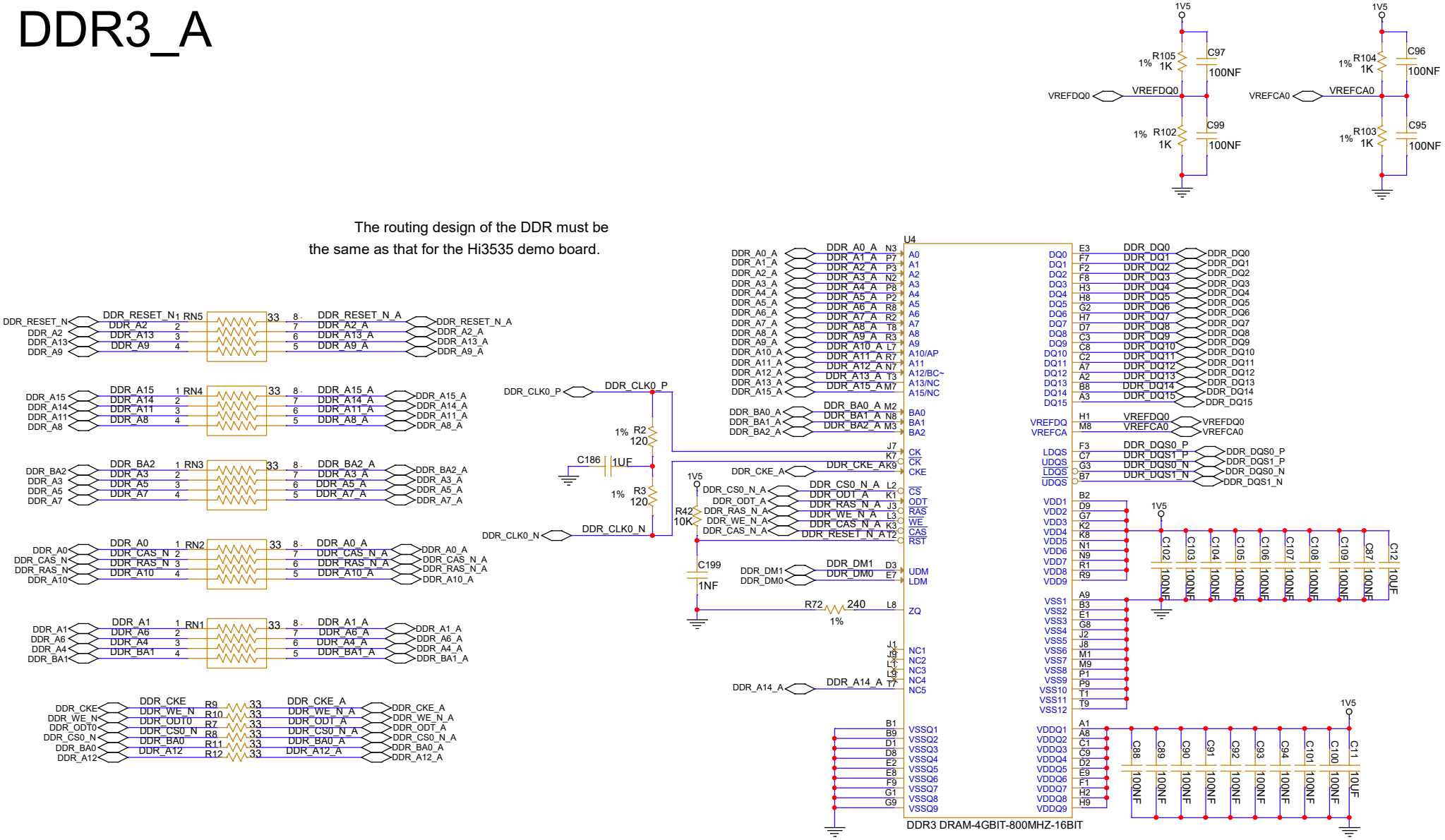
1000R@0.25A_0402

Temperature sampling

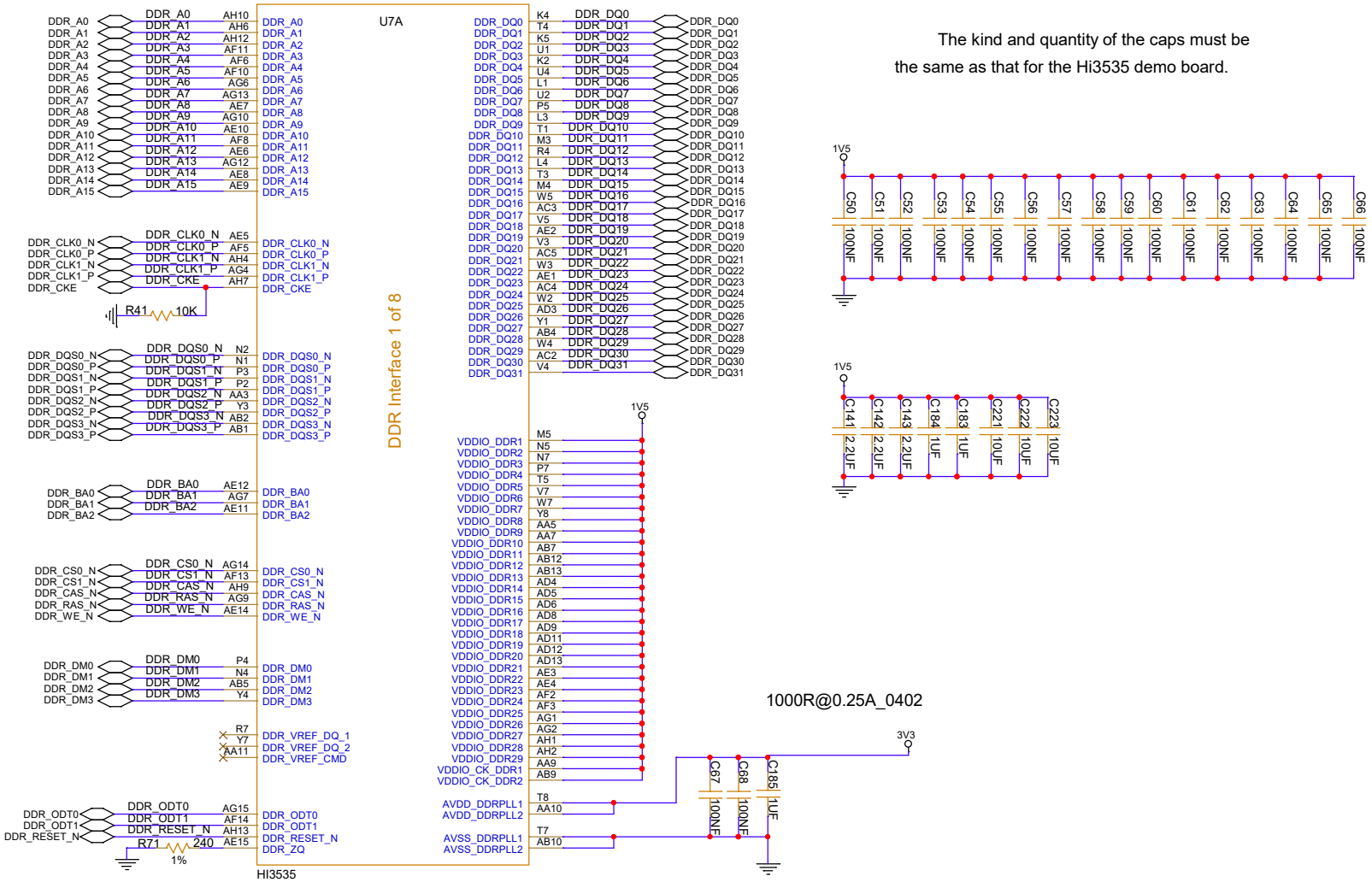
DDR3_B


1V5
1% R109 1K
C128 100NF
VREFDQ1
VREFDQ1
1% R106 1K
C129 100NF
1V5
1% R108 1K
C127 100NF
VREFCA1
VREFCA1
1% R107 1K
C126 100NF
The routing design of the DDR must be
the same as that for the Hi3535 demo board.
DDR_RESET_N
DDR_A2
DDR_A13
DDR_A9
DDR_RESET_N 8 RN9
DDR_A2
DDR_A13
DDR_A9
33
DDR_RESET_N_B
DDR_A2_B
DDR_A13_B
DDR_A9_B
DDR_A15
DDR_A14
DDR_A11
DDR_A8
DDR_A15 8 RN8
DDR_A14
DDR_A11
DDR_A8
DDR_A15_B
DDR_A14_B
DDR_A11_B
DDR_A8_B
DDR_BA2
DDR_A3
DDR_A5
DDR_A7
DDR_BA2 8 RN7
DDR_A3
DDR_A5
DDR_A7
DDR_BA2_B
DDR_A3_B
DDR_A5_B
DDR_A7_B
DDR_A0
DDR_CAS_N
DDR_RAS_N
DDR_A10
DDR_A0 8RN10
DDR_CAS_N
DDR_RAS_N
DDR_A10
DDR_A0_B
DDR_CAS_N_B
DDR_RAS_N_B
DDR_A10_B
DDR_A1
DDR_A6
DDR_A4
DDR_BA1
DDR_A1 8 RN6
DDR_A6
DDR_A4
DDR_BA1
DDR_A1_B
DDR_A6_B
DDR_A4_B
DDR_BA1_B
DDR_CKE
DDR_WE_N
DDR_ODT1
DDR_CS1_N
DDR_BA0
DDR_A12
DDR_CKE
DDR_WE_N
DDR_ODT1
DDR_CS1_N
DDR_BA0
DDR_A12
R14 33
R15 33
R13 33
R16 33
R18 33
R17 33
DDR_CKE_B
DDR_WE_N_B
DDR_ODT_B
DDR_CS1_N_B
DDR_BA0_B
DDR_A12_B
DDR_CLK1_P
DDR_CLK1_P
1% R4 120
C187 1UF
1% R5 120
DDR_CLK1_N
DDR_CLK1_N
1V5
R43 10K
C200 1NF
U5
DDR_A0_B
DDR_A1_B
DDR_A2_B
DDR_A3_B
DDR_A4_B
DDR_A5_B
DDR_A6_B
DDR_A7_B
DDR_A8_B
DDR_A9_B
DDR_A10_B
DDR_A11_B
DDR_A12_B
DDR_A13_B
DDR_A15_B
DDR_A0_B N3
DDR_A1_B P7
DDR_A2_B P3
DDR_A3_B N2
DDR_A4_B P8
DDR_A5_B P2
DDR_A6_B R8
DDR_A7_B R2
DDR_A8_B T8
DDR_A9_B R3
DDR_A10_B L7
DDR_A11_B R7
DDR_A12_B N7
DDR_A13_B T3
DDR_A15_B M7
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10/AP
A11
A12/BC~
A13/NC
A15/NC
DDR_BA0_B
DDR_BA1_B
DDR_BA2_B
DDR_BA0_B M2
DDR_BA1_B N8
DDR_BA2_B M3
BA0
BA1
BA2
J7
K7
CK
CK
DDR_CKE_B
DDR_CKE_B K9
CKE
DDR_CS0_N_B
DDR_ODT_B
DDR_RAS_N_B
DDR_WE_N_B
DDR_CAS_N_B
DDR_CS1_N_B L2
DDR_ODT_B K1
DDR_RAS_N_B J3
DDR_WE_N_B L3
DDR_CAS_N_B K3
DDR_RESET_N_B T2
CS
ODT
RAS
WE
CAS
RST
DDR_DM3
DDR_DM2
DDR_DM3 D3
DDR_DM2 E7
UDM
LDM
R73 240 1%
L8
ZQ
J1
J9
L1
L9
T7
NC1
NC2
NC3
NC4
NC5
DDR_A14_B
DDR_A14_B
B1
B9
D1
D8
E2
E8
F9
G1
G9
VSSQ1
VSSQ2
VSSQ3
VSSQ4
VSSQ5
VSSQ6
VSSQ7
VSSQ8
VSSQ9
DDR3 DRAM-4GBIT-800MHZ-16BIT
DQ0
DQ1
DQ2
DQ3
DQ4
DQ5
DQ6
DQ7
DQ8
DQ9
DQ10
DQ11
DQ12
DQ13
DQ14
DQ15
VREFDQ
VREFCA
LDQS
UDQS
LDQS
UDQS
VDD1
VDD2
VDD3
VDD4
VDD5
VDD6
VDD7
VDD8
VDD9
VSS1
VSS2
VSS3
VSS4
VSS5
VSS6
VSS7
VSS8
VSS9
VSS10
VSS11
VSS12
VDDQ1
VDDQ2
VDDQ3
VDDQ4
VDDQ5
VDDQ6
VDDQ7
VDDQ8
VDDQ9
E3 DDR_DQ16
F7 DDR_DQ17
F2 DDR_DQ18
F8 DDR_DQ19
H3 DDR_DQ20
H8 DDR_DQ21
G2 DDR_DQ22
H7 DDR_DQ23
D7 DDR_DQ24
C3 DDR_DQ25
C8 DDR_DQ26
C2 DDR_DQ27
A7 DDR_DQ28
A2 DDR_DQ29
B8 DDR_DQ30
A3 DDR_DQ31
H1 VREFDQ1
M8 VREFCA1
F3 DDR_DQS2_P
C7 DDR_DQS3_P
G3 DDR_DQS2_N
B7 DDR_DQS3_N
B2
D9
G7
K2
K8
N1
N9
R1
R9
A9
B3
E1
G8
J2
J8
M1
M9
P1
P9
T1
T9
A1
A8
C1
C9
D2
E9
F1
H2
H9
1V5
C110 100NF
C111 100NF
C112 100NF
C113 100NF
C114 100NF
C115 100NF
C116 100NF
C117 100NF
C118 100NF
C14 10UF
1V5
C119 100NF
C120 100NF
C121 100NF
C122 100NF
C123 100NF
C124 100NF
C125 100NF
C131 100NF
C130 100NF
C13 10UF

DDR3_A

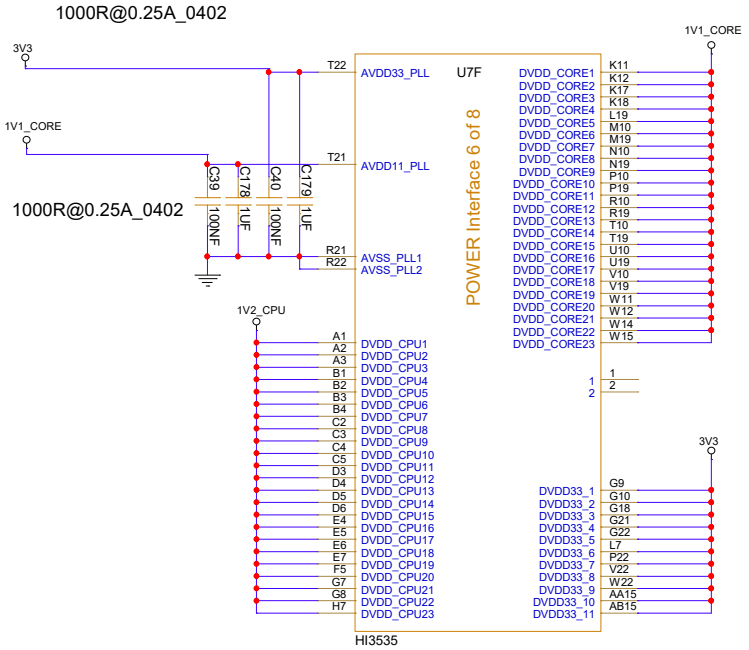

1V5
R105 1K 1%
C97 100NF
VREFDQ0
R102 1K 1%
C99 100NF
R104 1K 1%
C96 100NF
VREFCA0
R103 1K 1%
C95 100NF
The routing design of the DDR must be
the same as that for the Hi3535 demo board.
RN5
RN4
RN3
RN2
RN1
DDR_RESET_N
DDR_A2
DDR_A13
DDR_A9
DDR_A15
DDR_A14
DDR_A11
DDR_A8
DDR_BA2
DDR_A3
DDR_A5
DDR_A7
DDR_A0
DDR_CAS_N
DDR_RAS_N
DDR_A10
DDR_A1
DDR_A6
DDR_A4
DDR_BA1
DDR_CKE
DDR_WE_N
DDR_ODT0
DDR_CS0_N
DDR_BA0
DDR_A12
R9 R10 R7 R8 R11 R12
33
DDR_CLK0_P
DDR_CLK0_N
R2 120 1%
R3 120 1%
C186 1UF
R42 10K
C199 1NF
R72 240 1%
DDR_DM1
DDR_DM0
DDR_A14_A
DDR_CKE_A
DDR_CS0_N_A
DDR_ODT_A
DDR_RAS_N_A
DDR_WE_N_A
DDR_CAS_N_A
DDR_RESET_N_A
U4
DDR3 DRAM-4GBIT-800MHZ-16BIT
VREFDQ
VREFCA
LDQS
UDQS
VDD1 VDD2 VDD3 VDD4 VDD5 VDD6 VDD7 VDD8 VDD9
VSS1 VSS2 VSS3 VSS4 VSS5 VSS6 VSS7 VSS8 VSS9 VSS10 VSS11 VSS12
VDDQ1 VDDQ2 VDDQ3 VDDQ4 VDDQ5 VDDQ6 VDDQ7 VDDQ8 VDDQ9
VSSQ1 VSSQ2 VSSQ3 VSSQ4 VSSQ5 VSSQ6 VSSQ7 VSSQ8 VSSQ9
C102 C103 C104 C105 C106 C107 C108 C109 C87 C12
C88 C89 C90 C91 C92 C93 C94 C101 C100 C11
100NF 10UF

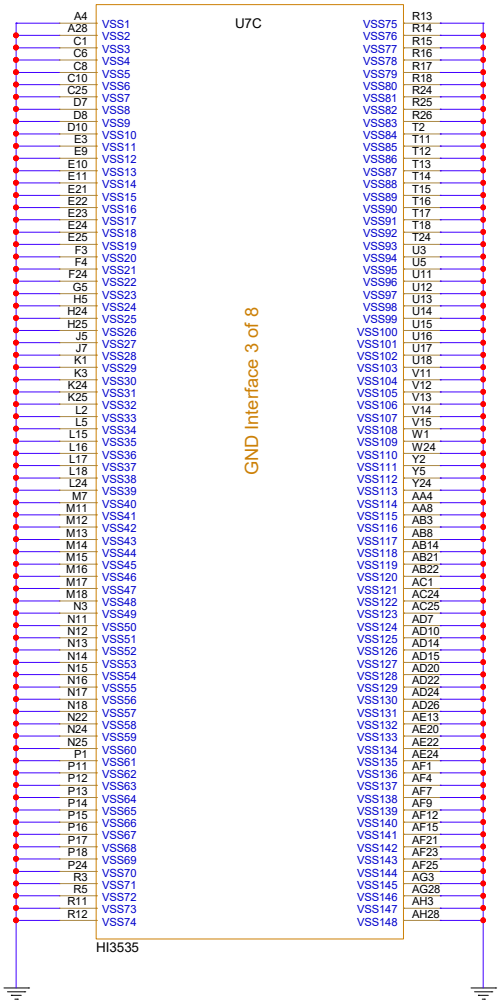
HI3535 DDR3



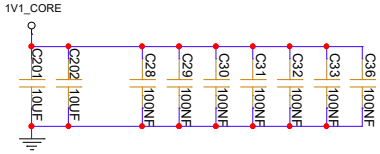
The kind and quantity of the caps must be the same as that for the Hi3535 demo board.

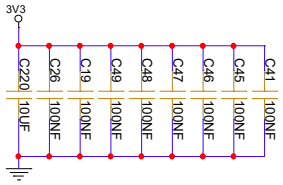
POWER & VSS

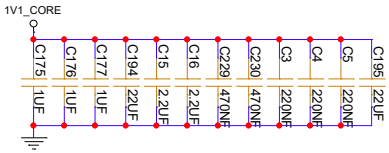


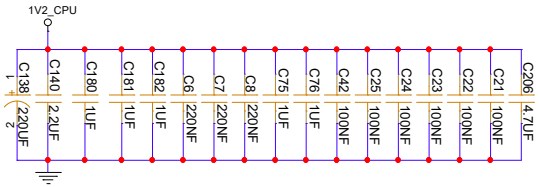


The kind and quantity of the caps must be the same as that for the Hi3535 demo board.







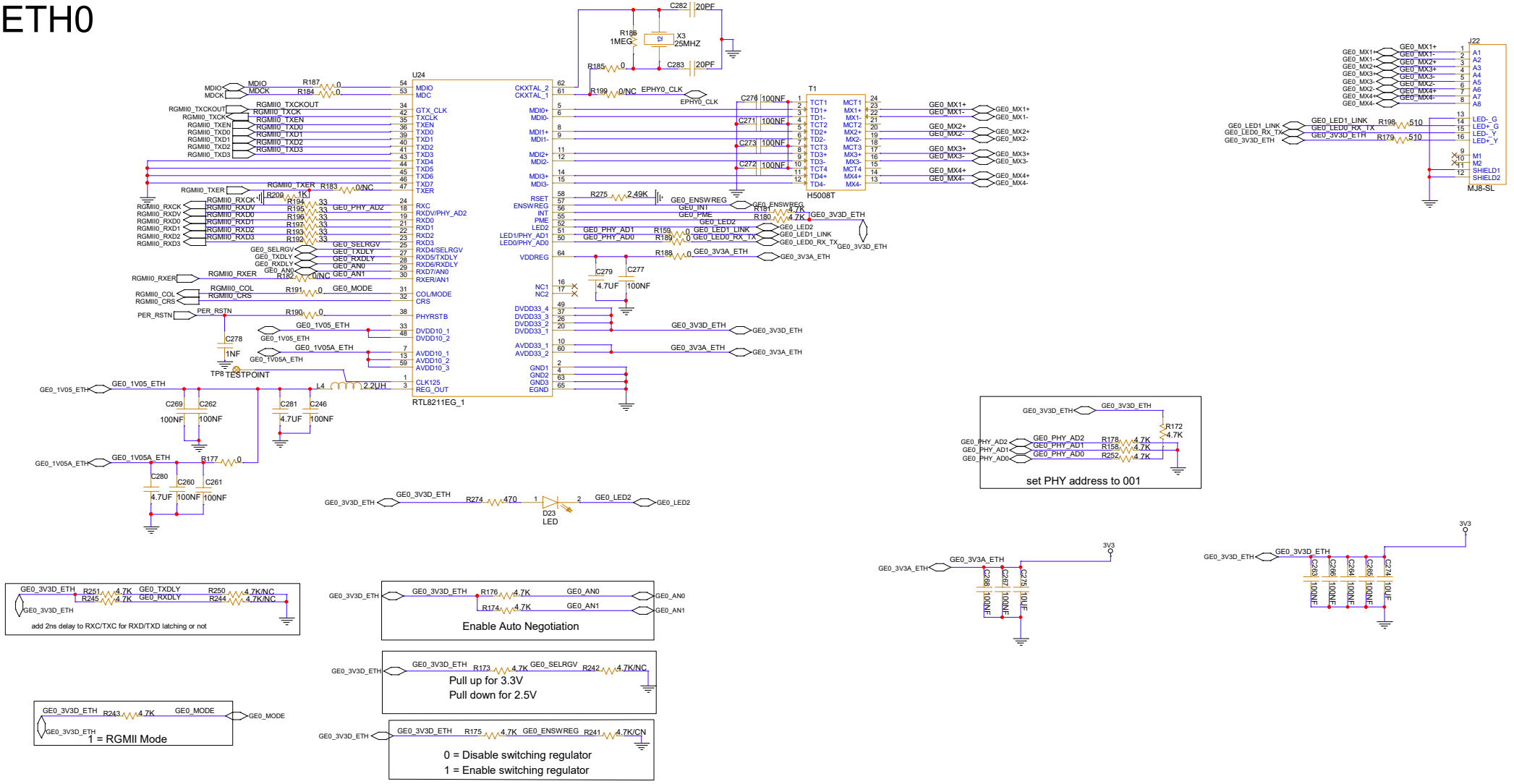


CHANGE LIST

2014.12.07 V1.0 schematic start

DC/DC 12V->DVDD_CPU 2Amax

ETH0


U24
RTL8211EG_1
T1
H5008T
J22
MJ8-SL
X3
25MHZ
C282 20PF
C283 20PF
R186 1MEG
R185 0
R199 0/NC
EPHY0_CLK
R187 0
R184 0
MDIO
MDCK
RGMII0_TXCKOUT
RGMII0_TXCK
RGMII0_TXEN
RGMII0_TXD0
RGMII0_TXD1
RGMII0_TXD2
RGMII0_TXD3
RGMII0_TXER
R183 0/NC
RGMII0_RXCK
RGMII0_RXDV
RGMII0_RXD0
RGMII0_RXD1
RGMII0_RXD2
RGMII0_RXD3
R209 1K
R194 33
R195 33
R196 33
R197 33
R193 33
R192 33
GE0_PHY_AD2
GE0_SELRGV
GE0_TXDLY
GE0_RXDLY
GE0_AN0
GE0_AN1
RGMII0_RXER
R182 0/NC
RGMII0_COL
RGMII0_CRS
R191 0
GE0_MODE
PER_RSTN
R190 0
C278 1NF
TP8 TESTPOINT
GE0_1V05_ETH
GE0_1V05A_ETH
L4 2.2UH
C269 100NF
C262 100NF
C281 4.7UF
C246 100NF
C280 4.7UF
C260 100NF
C261 100NF
R177 0
CXXTAL_2
CXXTAL_1
MDI0+
MDI0-
MDI1+
MDI1-
MDI2+
MDI2-
MDI3+
MDI3-
RSET
ENSWREG
INT
PME
LED2
LED1/PHY_AD1
LED0/PHY_AD0
VDDREG
NC1
NC2
DVDD33_4
DVDD33_3
DVDD33_2
DVDD33_1
AVDD33_1
AVDD33_2
GND1
GND2
GND3
EGND
MDIO
MDC
GTX_CLK
TXCLK
TXEN
TXD0
TXD1
TXD2
TXD3
TXD4
TXD5
TXD6
TXD7
TXER
RXC
RXDV/PHY_AD2
RXD0
RXD1
RXD2
RXD3
RXD4/SELRGV
RXD5/TXDLY
RXD6/RXDLY
RXD7/AN0
RXER/AN1
COL/MODE
CRS
PHYRSTB
DVDD10_1
DVDD10_2
AVDD10_1
AVDD10_2
AVDD10_3
CLK125
REG_OUT
R275 2.49K
GE0_ENSWREG
R181 4.7K
R180 4.7K
GE0_3V3D_ETH
GE0_3V3A_ETH
R159 0
R189 0
GE0_LED2
GE0_LED1_LINK
GE0_LED0_RX_TX
R188 0
C279 4.7UF
C277 100NF
C276 100NF
C271 100NF
C273 100NF
C272 100NF
TCT1
TD1+
TD1-
TCT2
TD2+
TD2-
TCT3
TD3+
TD3-
TCT4
TD4+
TD4-
MCT1
MX1+
MX1-
MCT2
MX2+
MX2-
MCT3
MX3+
MX3-
MCT4
MX4+
MX4-
GE0_MX1+
GE0_MX1-
GE0_MX2+
GE0_MX2-
GE0_MX3+
GE0_MX3-
GE0_MX4+
GE0_MX4-
GE0_LED1_LINK
GE0_LED0_RX_TX
GE0_3V3D_ETH
R198 510
R179 510
LED-_G
LED+_G
LED-_Y
LED+_Y
M1
M2
SHIELD1
SHIELD2
GE0_1V05_ETH
C269 100NF
C262 100NF
C281 4.7UF
C246 100NF
GE0_1V05A_ETH
C280 4.7UF
C260 100NF
C261 100NF
R177 0
GE0_3V3D_ETH
R274 470
D23
LED
GE0_LED2
GE0_3V3D_ETH
R172 4.7K
GE0_PHY_AD2
GE0_PHY_AD1
GE0_PHY_AD0
R178 4.7K
R158 4.7K
R252 4.7K
set PHY address to 001
GE0_3V3D_ETH
R251 4.7K
R245 4.7K
GE0_TXDLY
GE0_RXDLY
R250 4.7K/NC
R244 4.7K/NC
add 2ns delay to RXC/TXC for RXD/TXD latching or not
GE0_3V3D_ETH
R176 4.7K
R174 4.7K
GE0_AN0
GE0_AN1
Enable Auto Negotiation
GE0_3V3D_ETH
R173 4.7K
GE0_SELRGV
R242 4.7K/NC
Pull up for 3.3V
Pull down for 2.5V
GE0_3V3D_ETH
R243 4.7K
GE0_MODE
1 = RGMII Mode
GE0_3V3D_ETH
R175 4.7K
GE0_ENSWREG
R241 4.7K/CN
0 = Disable switching regulator
1 = Enable switching regulator
GE0_3V3A_ETH
3V3
C268 100NF
C267 100NF
C275 10UF
GE0_3V3D_ETH
3V3
C263 100NF
C266 100NF
C264 100NF
C265 100NF
C274 10UF